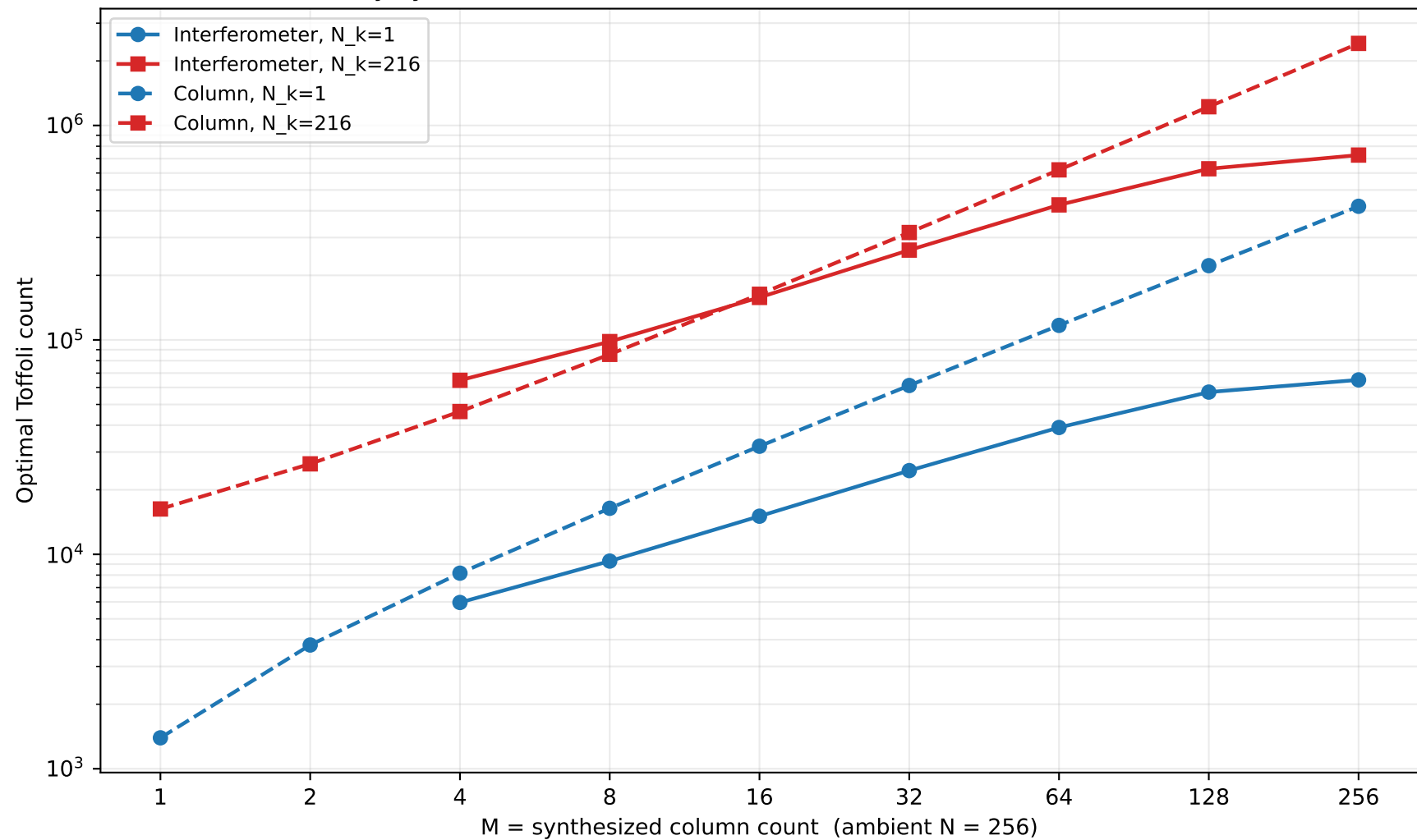


Isometry synthesis: interferometer vs column, Toffoli vs M (N=256, b=32)



Isometry synthesis optimal Toffoli: interferometer vs column (ambient N = 256, b = 32)

M	Interferometer N_k=1	Interferometer N_k=216	Column N_k=1	Column N_k=216
1	n/a (M<4)	n/a (M<4)	1,394	16,274
2	n/a (M<4)	n/a (M<4)	3,776	26,418
4	5,962	64,801	8,168	46,334
8	9,301	98,196	16,394	85,608
16	15,058	157,786	31,916	163,226
32	24,575	262,254	61,286	316,788
64	39,040	426,112	116,864	620,750
128	57,089	628,094	221,882	1,222,536
256	65,082	727,560	419,828	2,414,018